

Rotation and Composition Findings

12 February 2026

Questions

- Is the composed kernel rotation invariant?
- Why is composite accuracy lower than the regular all-kernel model, and how can we reduce the gap?

Rotation-Invariance

Composition is a weighted sum:

$$K_{\text{comp}} = \sum_i w_i K_i$$

This is not automatically rotation invariant. It becomes invariant only if the selected kernels and weights are rotationally balanced, or if the kernels are radially symmetric. Several kernel families used here are orientation-sensitive (e.g., anisotropic Gaussian, Gabor, HOG, LBP, GLCM), so the composite is expected to change under rotation.

Empirical checks on the saved composite kernel show large differences after 90°/180°/270° rotations.

Conclusion: the current composite kernel is not rotation invariant.

Non-Invariant But Still Robust

Even though the kernel itself is not rotation invariant, accuracy changes under 90° rotation were small:

- Regular model: $\Delta\text{AUC} = +0.0017$, $\Delta\text{ACC} = +0.0123$
- Composite model: $\Delta\text{AUC} = +0.0086$, $\Delta\text{ACC} = +0.0241$

Why accuracy stays stable:

- The response function `abs_max` is insensitive to sign and location of the strongest response, so many rotations yield similar feature magnitudes.
- Logistic regression relies on pooled response magnitudes rather than explicit orientation alignment.
- Many kernels capture texture/contrast statistics that are preserved under right-angle rotations.

Why Composite is Lower

Best current results (12 February 2026):

Model	Test AUC	Test ACC
Regular (separate kernels)	0.9376	0.8888
Composite (8 kernels)	0.9026	0.8299

Training-stage AUC/ACC: regular 0.9243/0.8615 vs composite 0.9233/0.8541.

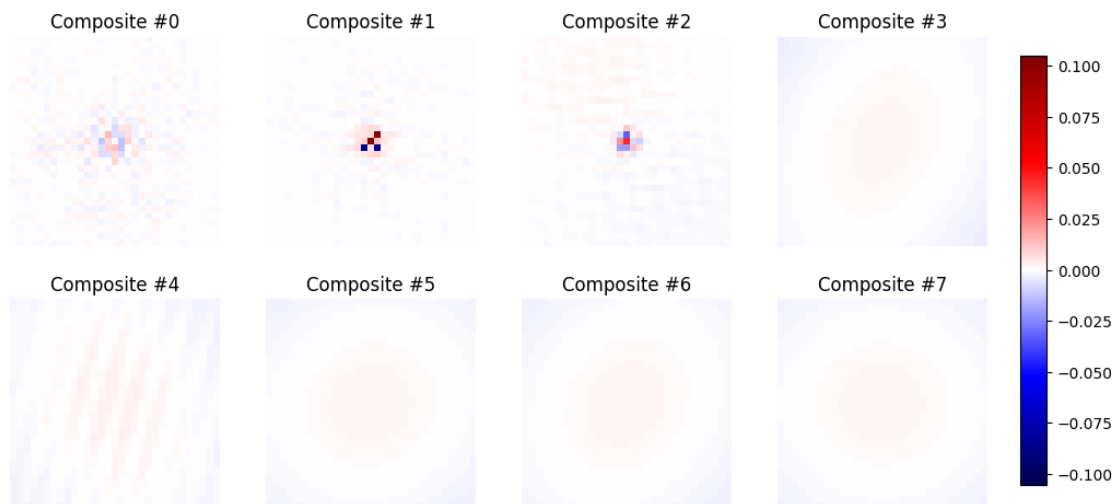


Figure 1: Composite kernel grid (8 composites).

Main factors lowering composite accuracy:

- **Feature collapse:** the regular model uses many kernel-response features; the composite reduces this to a small set, which compresses discriminative information.
- **Nonlinearity mismatch:** with `abs_max` aggregation, composing kernels before response is not equivalent to combining per-kernel responses after `abs_max`.
- **Heuristic weights:** composition weights are uniform or score-based, not optimized end-to-end for classification.

How to Reduce the Gap

- Use a **small bank of composites** (e.g., 8–32) instead of a single composite.
- Learn composition weights end-to-end with the classifier objective rather than using fixed heuristics.
- Make the response more linear (e.g., mean or sum pooling) so “compose then respond” is closer to “respond then combine.”
- If strict rotation invariance is required, tie weights across rotated kernel orbits; note this may reduce accuracy if orientation cues are informative.

Qualitative Evidence (Heatmap Pairs)

The following examples show side-by-side kernel and composite response visualizations for selected images. They illustrate how composition preserves some structure but can smooth or redistribute fine-grained responses.

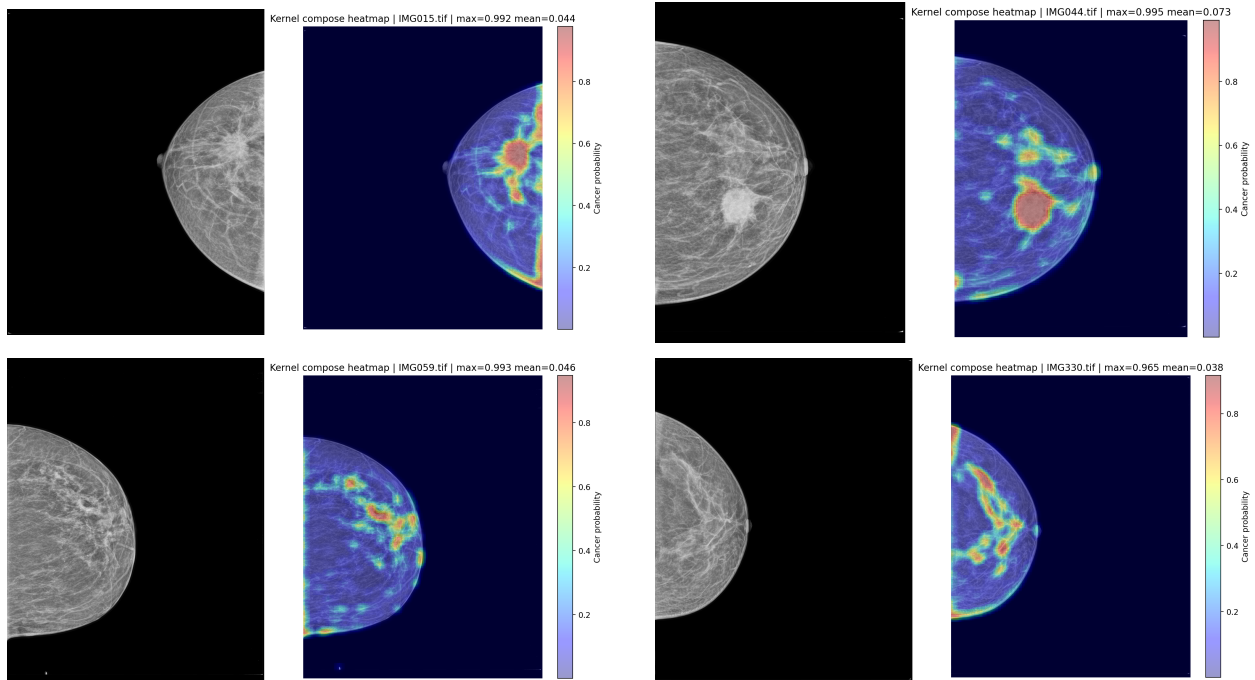


Figure 2: Kernel vs composite heatmap pairs for representative images.

Current Pipeline and Next Step

- Current: kernel generation is random across parameter ranges.
- Change: smarter sampling using QMC or Latin Hypercube.
- Current: normalization is applied to composed kernels; non-composed kernels are used without an extra normalization stage. Prior runs without extra normalization performed better.
- Change: test per-family normalization (different normalization per kernel type).

Conclusion

- The current composite kernel is not rotation invariant, yet accuracy remains stable under 90° rotations.
- Composite accuracy is lower mainly due to feature collapse and nonlinearity mismatch.
- Closing the gap is more realistic with multiple learned composites than with a single kernel.
- Larger composition pools can look “zoomed” due to cancellation and normalization effects.